PRODUCT USER MANUAL

For the two Baltic Sea Products:

Biogeochemical Analysis and Forecasting Product

BALTICSEA_ANALYSISFORECAST_BGC_003_007

&

Physical Analysis and Forecasting Product

BALTICSEA_ANALYSISFORECAST_PHY_003_006

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2022/06	All	New document now including both products: BALTICSEA_ANALYSISFORECAST_ BGC_003_007 BALTICSEA_ANALYSISFORECAST_ PHY_003_006	Anja Lindenthal, , Patrik Ljungemyr,	Vibeke Huess
1.1	2023/06	All	Updated to include the changes for the November 2023 system upgrades	Anja Lindenthal, Patrik Ljungemyr	V. Huess
1.2	2024/11	All	Updated to include the changes for the November 2024 system upgrades	V. Huess	V. Huess
1.3	2025/11		Slight update for the November 2025 production upgrade	Patrik Ljungemyr, Vibeke Huess	V. Huess

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
BAL	Baltic
CF	Climate Forecast (convention for NetCDF)
PC	Production Center
PU	Production Unit
MDS	Marine Data Store

DATA ACCESS

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Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

I INTRODUCTION

I.1 Summary

This document presents a detailed description of the two products:

- BALTICSEA_ANALYSISFORECAST_BGC_003_007
- BALTICSEA_ANALYSISFORECAST_PHY_003_006

with respect to available data services and how to access them, as well as how to use the files and services.

These two products contain the operational forecasts for the biogeochemical and the physical conditions in the Baltic Sea provided by the Copernicus Marine Service Baltic Monitoring and Forecasting Centre (BAL MFC). The two products are described together in this User Manual as the production is run as one online-coupled system.

The product BALTICSEA_ANALYSISFORECAST_PHY_003_006 is from November 2024 and onwards simulated with the BAL MFC-NEMO set up based on the open source ocean circulation model NEMO 4.2.1 that has been tuned and tested within the BAL MFC consortium to fit the needs of the Baltic Sea region. For details on product quality see the CMEMS-BAL-QUID-003-006 document available via the CMEMS on-line Catalogue.

The product BALTICSEA_ANALYSISFORECAST_BGC_003_007 is based on simulations with the BAL MFC-ERGOM version of the biogeochemical model ERGOM originally developed at IOW, Germany. The BAL MFC-ERGOM version has been developed further at BSH to include a silicate cycle and dissolved iron. The BAL MFC's set up of the NEMOv4.2.1 ocean model code has been extended to include a coupling to The Framework for Aquatic Biogeochemical Models (FABM; Bruggeman & Bolding, 2014) in the TOP module. The BAL MFC ERGOM version is hence designed to be coupled to NEMOv4.2.1 via FABM. For details of the quality of the ERGOM results see the CMEMS-BAL-QUID-003-007 document available via the CMEMS on-line Catalogue.

The BALTICSEA_ANALYSISFORECAST_BGC_003_007 product provides model forecasts for these data variable groups:

- Nutrients (phosphate, nitrate, ammonium, silicate and iron),
- Oxygen (dissolved oxygen and bottom oxygen),
- Plankton (chlorophyll-a, phytoplankton, zooplankton),
- Primary production (net primary production),
- Optical properties (light attenuation, Secchi depth and colored dissolved organic matters)
- Carbon (surface partial pressure of CO2, ph and dissolved inorganic carbon) and

- Hydrogen sulfide

All products are stored as 3D values with up to 56 levels, except for the Secchi depth, light attenuation, the surface partial CO2 pressure and the bottom dissolved oxygen, which are stored as 2D values.

Data are provided in different datasets:

- One dataset with daily mean averaged values of the above parameter list. The Secchi depth and the light attenuation are included as instantaneous values from noon UTC.
- One dataset with monthly mean averaged values of the above parameter list, except the Secchi depth and the light attenuation.
- One dataset with accumulated daily values of the net primary production.
- One dataset with static files with coordinate and bathymetry information.

The product BALTICSEA_ANALYSISFORECAST_PHY_003_006 contains model analysis and forecasts for the physical condition in the Baltic Sea. Data are served in different datasets:

- One dataset with hourly resolution for the parameters: sea surface height variations, sea ice concentrations and thickness at the surface and temperature, salinity and horizontal and vertical velocities for 56 vertical levels.
- One dataset with daily mean averaged values of the above parameter list
- One dataset with de-tided values (using a Doodson 39 hours filter) of sea surface height (using a Doodson 39 hours filter, applied for the midnight values).
- One dataset with de-tided values of surface horizontal velocities (using a Doodson 39 hours filter, applied for the midnight values).
- One dataset with monthly mean averaged values of the above parameter list
- One dataset with 15 minutes resolution for the sea surface height variations and surface currents.
- One dataset with static files: with coordinate, bathymetry and Mean Dynamic Topography information

I.2 History of changes

Date	Description of changes and impacted product
2020/12/15	BGC: Introduced as a new product in the CMEMS Catalogue. Supersedes the old product BALTICSEA_ANALYSIS_FORECAST_BIO_003_007 based on ERGOM forced with the HBM ocean model.

	PHY: Introduced as a new product in the CMEMS Catalogue. Supersedes the old product BALTICSEA_ANALYSIS_FORECAST_PHY_003_006 based on the HBM ocean model.
2021/12/14	On December 14 2021 addition of **static files.**
2022/11/29	BGC: - The new product variables phytoplankton, zooplankton, dissolved inorganic carbon, light attenuation and silicate are included. - Improved representation of surface dynamics of nitrate and phosphate concentrations due to parameterization of bioavailable fractions from nutrient loads. - Improved representation of nitrate, hypoxia in deep basins due to higher sinking velocity of detritus and lower remineralization rates for detritus and sediments. PHY: - Vertical sea water velocity is included - Temperature and salinity in-situ observations are assimilated into the model.
2023/11/29	BGC: The product is updated by these improvements in the forcing dataset: - The atmospheric deposition input data for NO3 and NH4 will be updated to use the EMEP's monthly dataset, instead of yearly. - The input for atmospheric CO2 concentrations will be updated to use a monthly dataset from the CAMS global greenhouse gas reanalysis, instead of a timeseries derived from NOAA data. PHY: The product is updated by these improvements: - the data assimilation system has been upgraded to a newer version developed by AWI, Germany - observations of ice concentration has been included in the data assimilation Further the data we include into our model domain at our open lateral boundaries in the North Sea will be changed to use a new dataset from the new NWS MFC consortium. Please, notice that the impact from this change has not been tested.
2024/11/26	BGC: - The model code, ERGOM is updated to a newer version developed at IOW, Germany. - Three new 3D prognostic variables included in the product files. - The coupling to the NEMO system is done by the FABM coupler. PHY: - The model code is upgraded from NEMOv4.0 to NEMOv4.2.1. Further one of the twice daily productions, the 00Z midnight run, is extended with 4 extra forecast days to deliver a 10 days forecast product. The 12Z

	production will keep the present 6 days forecast length.
2025/11/25	The forecast length will be extended and we will from November 2025 provide a new nine days forecast twice daily. This will impact our dataset structure and these 7 datasets will be retired by November 2025, as they will become obsolete: cmems_mod_bal_phy_anfc_7-10days_PT15M-i cmems_mod_bal_phy_anfc_7-10days_PT1H-i cmems_mod_bal_phy_anfc_7-10days_P1D-m cmems_mod_bal_phy-cur_anfc_detided-7-10days_P1D-m cmems_mod_bal_phy-ssh_anfc_detided-7-10days_P1D-m cmems_mod_bal_bgc_anfc_7-10days_P1D-m cmems_mod_bal_bgc-pp_anfc_7-10days_P1D-i Additional there will be a slight change in the production cycle with both twice daily productions being extended backwards in time to start with a 12 hour re-analysis before the actual nine days forecast start. This slight change in our production system is expected to improve the quality by including re-analyzed atmospheric forcing for these first 12 hours, and also allowing more observations to be ready for being included in our ocean data assimilation at production start. The files to be uploaded into the Catalogue from each production will therefore in addition to the files covering the new nine days forecast also include an overwriting of the data covering the 12 hours before the analysis time.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	BALTICSEA_ANALYSIS_BGC_003_007
Geographical coverage	9°E→ 30°E ; 53°N → 66°N
Variables	• Chlorophyll-a • Phytoplankton • Zooplankton • Nitrate • Phosphate • Ammonium • Silicate • Bottom Dissolved Oxygen • Bottom Oxygen • Dissolved Inorganic Carbon • Ph • Partial CO2 pressure at the surface • Secchi Depth • Light Attenuation • Net primary production • Iron • Colored Dissolved Organic Matter (CDOM) • Hydrogen sulfide
Product Type	Analysis and Forecast Near Real Time
Available time series	A new 9 days forecast is produced twice daily and updated into the

	catalogue at 10 UTC and 22 UTC, respectively. The data for the previous 12 hours, relative to analysis time, are overwritten by each production with a new calculated hindcast ; so in total a 228 hour long time series (12H hindcast + 9*24H forecast) are uploaded into the Marine Data Store twice daily. Model data are available for the latest two years (a running window).
Temporal resolution	Daily mean values. Accumulated daily values (only for the net primary production). Monthly mean values.
Target delivery time	New Forecast: twice daily at 10 UTC and 22 UTC
Horizontal resolution	App. 1 nautical miles: Delta Longitude: 1'40'' or 0.028 degrees Delta Latitude: 1' or 0.017 degrees
Number of vertical levels	Up to 56 levels.
Format	Netcdf & CF1.0

Table 1: BALM-MFC BGC Real Time products

Product Lines	BALTICSEA_ANALYSISFORECAST_PHY_003_006
Geographical coverage	9°E→ 30°E ; 53°N → 66°N
Variables	• Sea surface variations • Ice concentration and thickness • Temperature • Salinity • Horizontal velocities (eastward and northward) • Vertical velocities • Mixed Layer Depth (MLD) • Bottom Temperature • Bottom Salinity
Product Type	Analysis and Forecast Near Real Time
Available time series	A new 9 days forecast is produced twice daily and updated into the Marine Data Store at 10 UTC and 22 UTC, respectively. The data for the previous 12 hours, relative to analysis time, are

	overwritten by each production with a new calculated hindcast ; so in total a 228 hour long time series (12H hindcast + 9*24H forecast) are uploaded into the Marine Data Store twice daily. Model data are available for the latest two years (a running window).
Temporal resolution	15 minutes instantaneous data for sea level and surface currents Hourly instantaneous values Daily mean values De-tided daily men values (ssh and surface horizontal velocities. Using a Doodson 39 hours filter applied at the midnight values) Monthly mean values
Target delivery time	New Forecast: twice daily at 10 UTC and 22 UTC.
Horizontal resolution	App. 1 nautical miles: Delta Longitude: 1’40’’ or 0.028 degrees Delta Latitude: 1’ or 0.017 degrees
Number of vertical levels	Up to 56 levels
Format	Netcdf & CF1.0

Table 2: BALM-MFC PHY Real Time products

II.2 Details of datasets

BALTICSEA_ANALYSISFORECAST_BGC_003_007
Dataset: **cmems_mod_bal_bgc-pp_anfc_P1D-i** contains: daily accumulated values for the net primary production (nppv) Updated twice per day at 10 and 22 UTC. **cmems_mod_bal_bgc_anfc_P1D-m** contains: daily mean values (24 hours average) of all the below listed parameters (except nppv, zsd and kd) instantaneous values for the Secchi depth (zsd) and the light attenuation (kd) valid at noon UTC. Updated twice per day at 10 and 22 UTC . **cmems_mod_bal_bgc_anfc_P1M-m** contains: monthly mean values of all the below listed parameters (except nppv, zsd and kd) Updated once per month **Variable names in the NetCDF file and Unit, Long_name & Standard_name**
no3 [mmol /m3] Nitrate_Concentration **mole_concentration_of_nitrate_in_sea_water**
po4 [mmol /m3] Phosphate_Concentration **mole_concentration_of_phosphate_in_sea_water**
nh4 [mmol /m3] Ammonium_Concentration **mole_concentration_of_ammonium _in_sea_water**
si [mmol/m3] Silicate Concentration **mole_concentration_of_silicate_in_sea_water**
chl [mg /m3] Chlorophyll_a_Concentration **mass_concentration_of_chlorophyll_a_in_sea_water**
phyc [mmol/m3] Phytoplankton Concentration **mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water**
zooc [mmol/m3] Zooplankton Concentration **mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water**
o2 [mmol/m3] Dissolved_Oxygen_Concentration **mole_concentration_of_dissolved_molecular_oxygen_in_sea_water**
o2b [mmol /m3] Sea floor Dissolved_Oxygen_Concentration **mole_concentration_of_dissolved_molecular_oxygen_in_sea_water**

zsd [m] Secchi_depth **secchi_depth_of_sea_water** ***Notice: zsd is provided as instantaneous values valid at noon UTC and not available as daily mean and monthly mean values.***
kd [m-1] Light Attenuation **volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water** ***Notice: kd is provided as instantaneous values valid at noon UTC and not available as daily mean and monthly mean values.***
pH [1] sea_water_ph **sea_water_ph_reported_on_total_scale**
spCO2 [Pa] surface_partial_pressure_of_carbon_dioxide **surface_partial_pressure_of carbon_dioxide_in_sea_water**
dissic [mol/m3] Dissolved inorganic carbon Concentration **mole_concentration_of_dissolved_inorganic_carbon_in_sea_water**
nppv [mg /(m3 day)] net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume **net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water**
Fe [mmol/m3] **mole_concentration_of_dissolved_iron_in_sea_water**
cdom [mmol/m^3] **mole_concentration_of_colored_dissolved_organic_matter_expressed_as_carbon_in_sea_water**
h2s [mmol/m^3] **mole_concentration_of_hydrogen_sulfide_in_sea_water**
Dataset: **cmems_mod_bal_bgc_anfc_static:** **contains the below 2 files** **Variable names in the NetCDF file and Unit, Long_name & Standard_name**
File: BAL-MFC_003_007_coordinate.nc: **e1t [m]** Cell dimension along X axis **e2t [m]** Cell dimension along Y axis **e3t [m]** Cell dimension along Z axis **cell_thickness**
File: BAL-MFC_003_007_mask_bathy.nc: **mask [1]** Land-sea mask: 1 = sea ; 0 = land **sea_binary_mask**

deptho [m] Bathymetry **sea_floor_depth_below_geoid** **deptho_lev [1]** Model level number at sea floor **model_level_number_at_sea_floor**

Table 3: list of the datasets and variable names and unit for the BALTICSEA_ANALYSISFORECAST_BGC_003_007 product

BALTICSEA_ANALYSISFORECAST_PHY_003_006
Dataset: **cmems_mod_bal_phy_anfc_PT1H-i** contains: hourly instantaneous values and is updated twice per day at 10 and 22 UTC. **cmems_mod_bal_phy_anfc_P1D-m** contains: daily mean values (24 hours average) updated twice per day at 10 and 22 UTC. **cmems_mod_bal_phy_anfc_P1M-m** contains: monthly mean values updated once per month **cmems_mod_bal_phy_anfc_PT15M-i** contains: 15 minutes sea level and surface currents updated twice per day at 10 and 22 UTC. **cmems_mod_bal_phy-ssh_anfc_detided_P1D-m** de-tided daily means for the sea level updated twice per day at 10 and 22 UTC. Notice: only data for forecastlength-1 day are available for the de-tided values due to the 39-hours centered window. **cmems_mod_bal_phy-cur_anfc_detided_P1D-m** contains: de-tided daily means for the surface horizontal velocities updated twice per day at 10 and 22 UTC. Notice: only data for forecastlength-1 day are available for the de-tided values due to the 39-hours centered window. **Variable name in the NetCDF file and Unit, Long_name & Standard_name**
sla [m] Sea level anomalies **sea_surface_height_above_sea_level**
u0 [m/s] Eastward current **eastward_sea_water_velocity**
v0 [m/s] Northward current **northward_sea_water_velocity**
w0 [m/s] Upward vertical velocity **upward_sea_water_velocity**
so [1e-3] Salinity **sea_water_salinity**
thetao [degree_C] Potential temperature **sea_water_potential_temperature**
sithick [m] Sea ice thickness **sea_ice_thickness**

siconc [0, 1] Sea ice cover **sea_ice_area_fraction**
mlotst [m] Ocean mixed layer thickness defined by density (as in de Boyer Montegut, 2004) **ocean_mixed_layer_thickness_defined_by_sigma_theta**
bottomT [degree_C] Sea floor potential temperature **sea_water_potential_temperature_at_sea_floor**
sob [1e-3] Sea water salinity at sea floor **sea_water_salinity**
Daily de-tided sea level: **zos_detided** [m] **sea_surface_height_above_geoid_assuming_no_tide**
Daily de-tided surface horizontal velocities: **uo_detided** [m/s] **northward_sea_water_velocity_assuming_no_tide;** **vo_detided** [m/s] **eastward_sea_water_velocity_assuming_no_tide**
Dataset: **cmems_mod_bal_phy_anfc_static** **contains the below 3 files** **Variable names in the NetCDF file and Unit, Long_name & Standard_name**
File: BAL-MFC_003_006_coordinate.nc: **e1t [m]** Cell dimension along X axis **e2t [m]** Cell dimension along Y axis **e3t [m]** Cell dimension along Z axis **cell_thickness**
File: BAL-MFC_003_006_mask_bathy.nc: **mask [1]** Land-sea mask: 1 = sea ; 0 = land **sea_binary_mask** **deptho [m]** Bathymetry

sea_floor_depth_below_geoid **deptho_lev [1]** Model level number at sea floor **model_level_number_at_sea_floor**
File: BAL-MFC_003_006_mdt.nc: **mdt [m]** Mean dynamic topography **sea_surface_height_above_geoid**

Table4: list of the datasets and variable names and unit for the BALTICSEA_ANALYSISFORECAST_PHY_003_006 product

II.3 Production System Description

The Near Real Time (NRT) BAL MFC model system consists of four main parts, see Figure 1. The physical model NEMO, the Biogeochemical model ERGOM, the wave model WAM and the data assimilation system PDAF. NEMO and ERGOM are online-coupled and use PDAF to assimilate SST, T/S profiles and sea ice concentration. The NEMO-ERGOM-PDAF system uses Stokes drift data from the WAM simulation and WAM reads in ice concentration and sea surface currents from the NEMO production. The full NEMO-ERGOM system runs at a native resolution of 1 nautical mile and all models use the same bathymetry based on the GEBCO v2014 data.

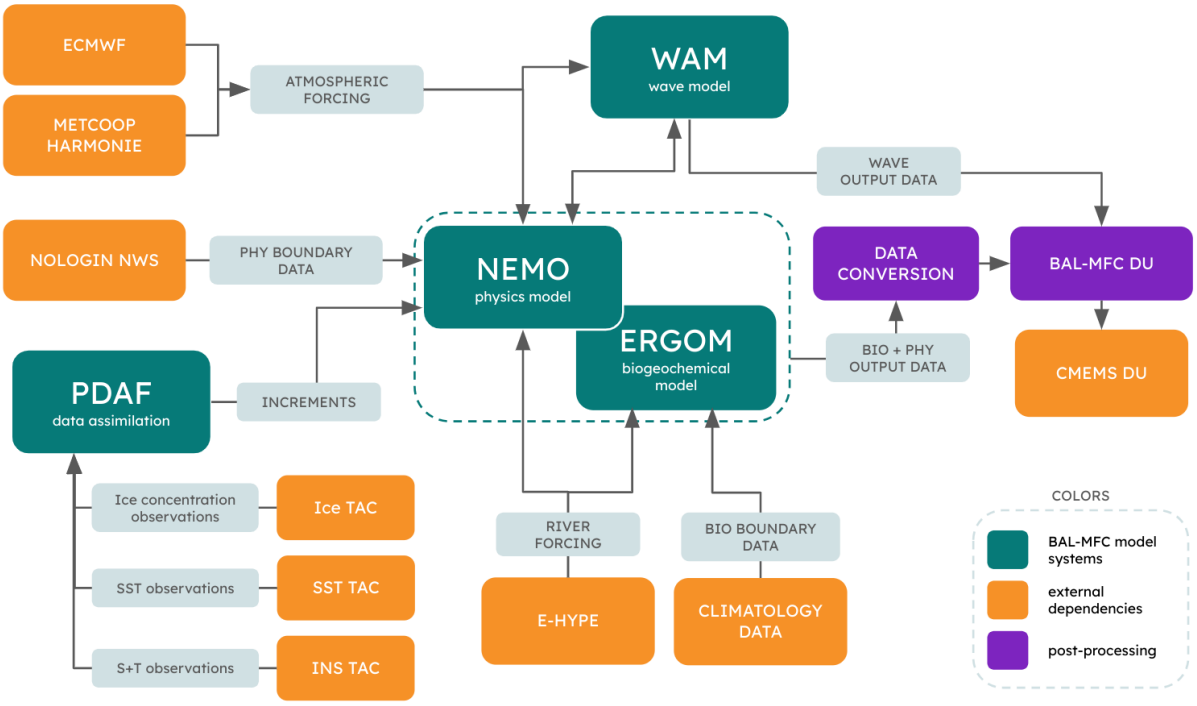


Figure 1 A system overview of the BAL MFC model system

NEMO

The product BALTICSEA_ANALYSISFORECAST_PHY_003_006 available from November 2024 (EIS202411 version) and onwards is simulated with the BAL MFC-NEMO set up based on the open source ocean circulation model NEMO 4.2.1 that has been tuned and tested within the BAL MFC consortium to fit the needs of the Baltic Sea region. For details about the product quality see the CMEMS-BAL-QUID-003-006 document available via the CMEMS on-line Catalogue.

Changes in the NEMO code since the previous version

Compared to the EIS202311 version the following changes have been done to the NEMO code:

- The NEMO code was upgraded from NEMOv4.0.4 to NEMOv4.2.1
- The number of ice-classes in the ice model is changed from 5 to 1 to better suit the data assimilation which only affects one ice class
- The chla- value in the standard NEMO set up is updated to use salinity scaled values. Compared to the use of spatial varying constants or a west/east value field the best results were achieved when using a salinity scaled chla-value.
- In the nemo namelist we now use the bulk formula: `ln_ECMWF = .true.` to be compatible with the met forcing used. Test runs showed that this setting gave smaller cold SST bias.
- The Smagorinsky value is updated to a 2D field tuned for better SSH and inflow results.

ERGOM

The product BALTICSEA_ANALYSISFORECAST_BGC_003_007 is based on simulations with the BAL MFC-ERGOM version of the biogeochemical model ERGOM originally developed at IOW, Germany. The BAL MFC-ERGOM version has been developed further at BSH to include a silicate cycle and dissolved iron. The BAL MFC's set up of the NEMOv4.2.1 ocean model code has been extended to include a coupling to The Framework for Aquatic Biogeochemical Models (FABM; Bruggeman & Bolding, 2014) in the TOP module. The BAL MFC ERGOM version is hence designed to be coupled to NEMOv4.2.1 via FABM. The ecosystem model describes the basic nutrient and carbon cycles through 20 pelagic state variables. The sediment is not vertically resolved and consists of two organic nutrient components. Figure 2 shows a schematic overview of all the variables and processes of the BAL MFC ERGOM version. For details on the ERGOM biogeochemical module, please see Maar et al. 2011 and Neumann et al. 2021. For details of the quality of the ERGOM results see CMEMS-BAL-QUID-003-007, which is available via the CMEMS on-line Catalogue.

Changes in the ERGOM code since previous version

Compared to the EIS202311 version the following changes have been made to the ERGOM code:

- Coupled with FABM (Bruggeman & Bolding, 2014)
- The ERGOM code is now based on the latest version developed at IOW, described in Neuman et a. 2021, and has additionally been extended with a silicate cycle and dissolved iron
- The radiation model is the two-band light model used in the General Ocean Turbulence Model (GOTM), provided by and coupled via the FABM coupler (https://doi.org/10.5281/zenodo.3774497)
- New variables in the product: colored dissolved organic matter (CDOM), hydrogen sulfide (H2S), dissolved iron (Fe3+)

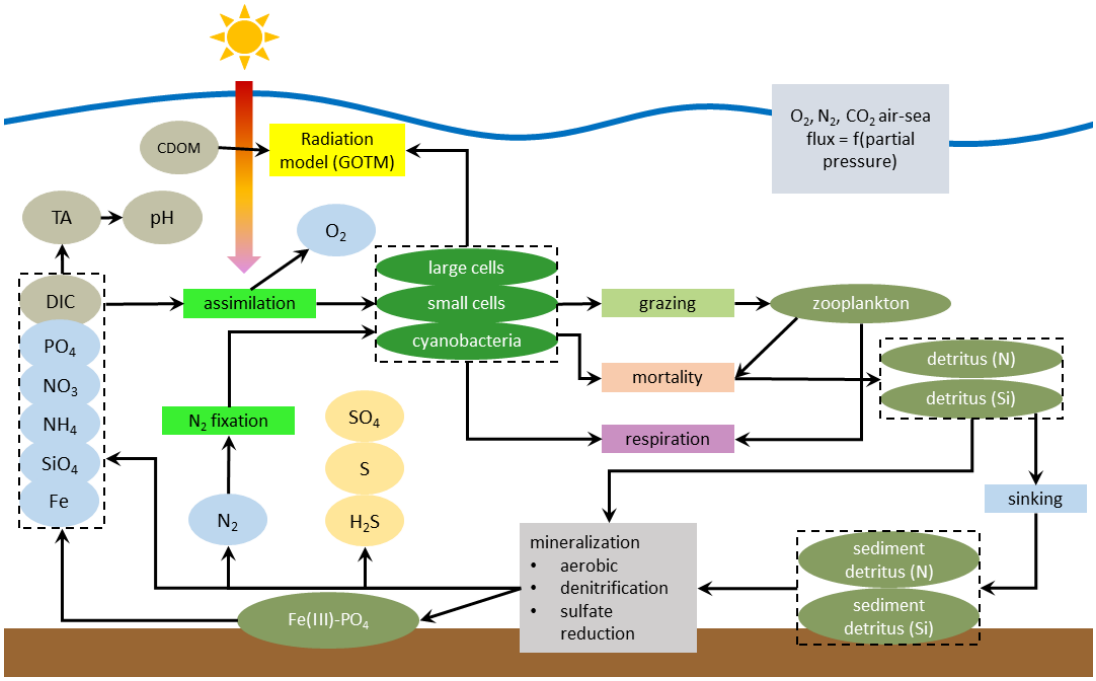


Figure 2 Schematic overview of variables and processes in ERGOM, the biogeochemical component of the BAL MFC model system.

II.3.1 Boundary forcing

The coupled model sytem NEMO-ERGOM is, at the two open boundaries (Orkney Island, UK - Norway & English Channel) forced with data from CMEMS' Atlantic North West Shelf forecast system (NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013). We use the available data from the NWS system at our open boundaries, and for the last day(s) several parameters in the boundary data are kept constant and tides are included from another source. This is done to make sure a stable production and to be able to deliver our forecast twice a day at reasonable hours. It should be noted that these product files (NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013) that we use as boundary data has changed during the years; and that it will change production system again from November 2025. However, we will adjust the input sea level reference level at our open lateral boundaries to keep the same reference level in our product; and there will be no changes in our Mean Dynamic Topography due to this change in the input forcing data.

For the biogeochemical model ERGOM interpolated data from ICES for DIN, DIP, silicate and oxygen are used. Zoo- and phytoplankton data are based on ICES observations of chlorophyll-a (chlorophyll to plankton biomass). The prognostic parameters of the carbonate system Total Alkalinity (TA) and Dissolved Inorganic Carbon (DIC) are based on Wolf-Gladrow et al. (2007) and results of the biogeochemical model ECOHAM (Lorkowski et al., 2012).

II.3.2 River and nutrients run-off:

Daily real time data from the HYPE forecast system from SMHI are used for river discharge, temperature and nutrients. The nutrient loads are converted into concentrations of nitrogen (inorganic and total) and phosphorus (soluble reactive and total). River loads of silicate, dissolved inorganic carbon and alkalinity are based on Pätsch and Lenhart (2012) and the Baltic database. River loads of CDOM are based on data from Neuman et al. 2021.

II.3.3 Surface forcing

Data of atmospheric deposition of oxidised and reduced nitrogen are downloaded from the EMEP database (https://emep.int/mscw/mscw_moddata.html). Both datasets are based on gridded monthly depositions for the Baltic and North Sea regions.

Monthly means of atmospheric CO_2 are read in from 2D fields generated from CAMS global greenhouse gas reanalysis product
(EGG4): https://ads.atmosphere.copernicus.eu/cdsapp#!/dataset/cams-global-ghg-reanalysis-egg4-monthly?tab=form

The atmospheric forcing is provided from the MetCoOp-HARMONIE 2.5 km system for the first 2.5 days, and ECMWF 11km forcing for the following days.

II.3.4 Data assimilation

The data assimilation method used in the NRT system is based on PDAF_OMI (Observation module infrastructure), an extension to PDAF, to provide a defined structure for observation operators and observation handling. The filter LESTKF is used. We have developed the observation interface for temperature profiles, salinity profiles, SST and ice concentration variables within PDAF_OMI. This, indeed, takes advantages of parallelism using MPI and also netCDF4 parallelism functionality, i.e. multiples cores can read one netCDF file in parallel mode. PDAF_OMI also has an ability to take into account only sea points, wet points, which reduces computational time and memory consumption. We use the univariate mode when reading observations and creating increments. But for the first guess and ensemble we read several variables. We use a spatially-varying localisation radius in PDA_OMI to limit the influence of observation on model field within a certain radius, depending on the geographical location. By default, PDAF_OMI only supports the constant localisation radius. We have implemented a two dimensional localisation radius by introducing a netcdf file to be read by PDAF. This is only used for the temperature and salinity profile assimilation. Upon on our findings during the validation and the tuning process of PDAF, however, we have decided to use a constant localisation radius for the sea ice concentration assimilation.

Observation files are interpolated to the BAL MFC NEMO grid as a preprocessing part. When doing assimilation for SST, we set the increment file for temperature to zero when there is sea ice. Also, for the variables, temperature, salt and SST, if the difference between the first guess estimate and the observation is more than 2, we mask those observation before running

PDAF.

During the summer time and ice free season, we assimilate only temperature, salinity and SST.

The observations used for the data assimilation are:

- **SST** - The Copernicus Marine Service L3 SST product for the Baltic Sea (https://doi.org/10.48670/moi-00154) provides daily night satellite observations centered around midnight
- **T & S profiles -** NRT profile observation data available at SMHI-database aggregated for 24 hours
- **Ice concentration (ICONC) –** FMI-BAL-SEAICE_CONC-L4-NRT-OBS (https://doi.org/10.48670/moi-00132)

II.3.5 Description of the Mean Dynamic Topography data

The mean dynamic topography "MDT" (available in file ***BAL-MFC_003_006_mdt.nc***) is calculated by taking the time mean of ssh for each grid point in the result files from a two-year run in production mode with data assimilation, and is updated for every production upgrade.

II.3.6 The runtime schedule

The production of the online coupled ocean-biogeochemical model system for providing the forecast product for the physical and biogeochemical conditions in the Baltic Sea is performed at SMHI (the Swedish Meteorological and Hydrological Institute), Sweden with two production cycles per day providing a new re-analysis for the previous 12 hours relative to analysis time and then followed by a nine days forecast. The data assimilation performed follows this schedule:

<u>For the 00Z analysis run:</u>

The run starts at 12Z the previous day

Data assimilation is applied at model start with:

- satellite observations of the Sea Ice Concentration (during winter time) using product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004 that is available at 15 UTC
- in situ observations of temperature and salinity profiles (during summer period: June 1st – December 1st)

For the 12Z analysis run:

The run starts at 0Z the same day

Data assimilation is applied at model start with:

- satellite observations of the sea surface temperature using product

  SST_BAL_SST_L3S_NRT_OBSERVATIONS_010_032) available at 6:30 UTC.

- in situ observations of temperature and salinity profiles (during winter period:

  December 1st – June 1st)

The ocean model run starts the 00z and 12Z productions when data from the weather model MetCOOP-HARMONIE at SMHI has finished the 00Z or 12Z forecast, respectively. The forecast from the MetCOOP HARMONIE does not cover the full North Sea area, and is therefore mixed with ECMWF forecast to get the full area coverage. Further, the atmospheric forcing consist of the control member of the high resolution MetCOOP/HARMONIE system for the first 66 hours and ECMWF deterministic forecast thereafter. The MetCOOP/HARMONIE wind forcing has 1 h temporal resolution; the ECMWF forecast has varying temporal resolution: for forecast 67 h to 90 h WAM uses hourly wind forcing from ECMWF, then resolution decreases to 3-hourly for forecast 93 h to 144 h, and then to 6-hourly for the last 3 days in our 9 days forecast.

The Baltic physical and biogeochemical product are updated twice per day into the Copernicus Marine Data Store at 10 UTC and 22 UTC, at the latest, for the 00Z UTC and 12Z UTC analyses, respectively.

Please notice that the run used for the 2 years validation run (2020-11-01 until 2022-10 -31) used for estimating the product quality in the QUID report, was made with 24H runs between the analysis. During wintertime we switched between two analyses types every second day:

During non-ice season: SST observations and S&T -profiles observations every day

During ice season:

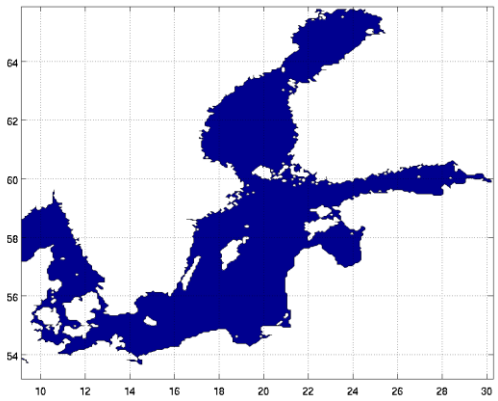
a) SST observations, T & S profiles every second day

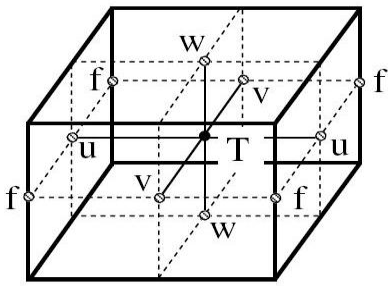
b) Ice concentration observations every second day

II.3.7 Summary of the operational model setups

Domain, resolution and grid

Geographic coverage: North Sea – Baltic Sea **:** 53°N - 66°N, 9°E - 30°E





The horizontal grid step is regular in latitude and longitude and is of approximately 1 nautical miles (lat: 0.017 degrees; lon: 0.028 degrees). The grid is a staggered Arakawa C-grid (see figure 1). The dataset of the scalar variables (temperature, salinity, sea level, ice, and biogeochemical variables) are centred in the T point of the grid; at the grid cell node. The velocity components are located half-way between adjacent grid cell nodes on the face of the grid cell in the respective direction. The velocities are de-staggered/interpolated to the T point in the delivered products.

Bathymetry	GEBCO at 1 nautical mile horizontal resolution
Vertical grid	The product is delivered at the models native grid, with up to 56 vertical levels: levels [m]: 0.50, 1.52, 2.55, 3.60, 4.68, 5.80, 6.96, 8.17, 9.45, 10.81, 12.27, 13.86, 15.60, 17.53, 19.70, 22.15, 24.94, 28.15, 31.86, 36.15, 41.13, 46.91, 53.59, 61.28, 70.08, 80.07, 91.31, 103.82, 117.62, 132.67, 148.90, 166.25, 184.60, 203.86, 223.92, 244.66, 265.99, 287.81, 310.05, 332.64, 355.51, 378.61, 401.91, 425.36, 448.93, 472.61, 496.36, 520.19, 544.06, 567.98, 591.94, 615.92, 639.92, 663.94, 687.98, 712.02
Model Version **Production units** **Period of validation**	▪ NEMO 4.2.1 – ERGOM_v202405 ▪ SMHI, Sweden ▪ 01.11.2020–31.10.2022
Assimilation scheme	PDAF LESTKF univariate for SST, sea ice concentration & T/S profiles

Assimilated observations	SST_BAL_SST_L3S_NRT_OBSERVATIONS_010_032 SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004 S & T profiles from SMHI databases in NRT forecast production. For the validation run a combination of ICES and SHARK-database is used.
Initial conditions (spin-up process)	For NEMO: Spin up started from January 1st 2020 with a restart field from the BALTICSEA_MULTIYEAR_PHY_003_011 product.
Surface forcing and boundary forcing	▪ Interpolated ICES data for DIN, DIP, silicate and oxygen. Zoo- and phytoplankton data are based on ICES observation of chlorophyll-a (chlorophyll to plankton biomass.) ▪ The calibration period used data from the MetCoOP HARMONIE 2.5km system. Since this data do not cover the full model domain it is extended by using ECMWF 11 km forcing. ▪ The forecast production is produced with forcing from the MetCoOp -HARMONIE 2.5 km system for the first 2.5 days, and the available ECMWF 11km forcing for the following days. ▪ The ocean model NEMO is, at the two open boundaries, forced with data from CMEMS Atlantic North West Shelf forecast system (NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013) for the available days of the forecast. For the last day(s) several parameters in the boundary data are kept constant and tides are included from another source. This is done to make sure a stable production and to be able to deliver our forecast twice a day at reasonable hours. ▪ Reading in Stokes drift from BAL MFC WAM-output into NEMO.
River run-off	Data from the operational hydrological model E-HYPE of SMHI.
NEMO-ERGOM coupling	The biogeochemical model is online coupled to the physical model through the FABM coupler.

II.4 Grid

The product files are on a regular lat-lon grid and all values are delivered at the Tracer-point (i.e. the 3D velocities are interpolated to the Tracer-points).

II.5 Processing information

II.5.1 Update Time

The BALTICSEA_ANALYSISFORECAST_PHY_003_006 product is updated twice daily in the Marine Service data catalogue at 10 UTC and 22 UTC.

The BALTICSEA_ANALYSISFORECAST_BGC_003_007 product is updated twice daily in the Marine Service data catalogue at 10 UTC and 22 UTC.

II.5.2 Temporal extent of analysis and forecast stored on delivery mechanism

For the twice daily forecast production the temporal extent is 9 days. Please notice that the de-tided values are only available for "forecast length minus one day", due to the 39-hour window used in the de-tide process. Data from the latest 2 years are available in the Marine Data Store for download (a running 2 years window).

II.5.3 Time averaging

Different time resolutions are provided for the two products:

15-minutes instantaneous: instantaneous values from every 15 min centered at hh:15min, hh:30min, hh:45min and hh:00min.

hourly instantaneous: instantaneous values once per hour from hh:00

daily accumulated values: values from every timestep is added during 24 hours

daily average values: daily mean values centered at noon (12:00 am)

monthly average values: monthly means over the calendar month (first to last day of the month).

De-tided values: doodson filtered hourly instantaneous values from a 39 hours window.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: *"How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"*

IV.1 File names

BALTICSEA_ANALYSISFORECAST_BGC_003_007: The files are in NetCDF and are named:

- For the daily mean values:
  BAL-ERGOM_BGC-DailyMeans-"ccyymmdd".nc
  Each file contains 1 timestamp for the calendar day.

- For the monthly mean values:
  BAL-ERGOM_BGC-MonthlyMeans-"ccyymm".nc
  Each file contains 1 timestamp for the calendar month.

- For the accumulated daily values:
  BAL-ERGOM_BGC-AccumulatedDaily-'ccyymmdd".nc
  Each file contains 1 timestamp for the calendar day.

BALTICSEA_ANALYSISFORECAST_PHY_003_006: The files are in NetCDF and are named:

- For the hourly instantaneous values:
  BAL-NEMO_PHY -"ccyymmddhh".nc
  Each file contains 12 hours of data. The timestamp indicate the start time in that file.

- For the daily mean values:
  BAL- NEMO_PHY -DailyMeans-"ccyymmdd".nc
  Each file contains 1 timestamp for the calendar day.

- For the de-tided values:
  BAL- NEMO_PHY –detided_ssh-"ccyymmdd".nc
  BAL- NEMO_PHY –detided_cur-"ccyymmdd".nc
  Each file contains 1 timestamp for the calendar day.

- For the monthly mean values:
  BAL- NEMO_PHY-MonthlyMeans-"ccyymm".nc
  Each file contains 1 timestamp for the calendar month.

- For the 15 minutes values:
  BAL- NEMO_PHY-15minutes-"ccyymmddhh".nc

Each file contains 48 timestamps (12 hours). The timestamp indicate the start time in that file.

IV.2 Other information: mean centre of Products, land mask value, missing value

Fill and land value: -999

All files are in NetCDF4 format.

IV.3 File size

BALTICSEA_ANALYSISFORECAST_BGC_003_007:

Daily accumulated files:	11 Mb/file
Daily mean files:	130 Mb/file
Monthly mean files:	app. 130 Mb/file

BALTICSEA_ANALYSISFORECAST_PHY_003_006:

15 minutes surface files:	71 Mb/file
Hourly files:	645 Mb/file
Daily mean files:	55 Mb/file
Detided ssh files :	2.3 Mb/file
Detided cur files:	4.6 Mb/file
Monthly mean files:	55 Mb/file

IV.4 Structure of file

Examples of the header of output NetCDF files are inserted in VI

V REFERENCES

Quality Information Document

CMEMS-BAL-QUID-003-006.pdf (copernicus.eu)

CMEMS-BAL-QUID-003-007.pdf (copernicus.eu)

Bruggeman, J., Bolding, K., 2014. A general framework for aquatic biogeochemical models. Environmental Modelling & Software 61: 249–265. DOI: 10.1016/j.envsoft.2014.04.002

Neumann, T., S. Koponen, J. Attila, C. Brockmann, K. Kallio, M. Kervinen, C. Mazeran, D. Müller, P. Philipson, S. Thulin, S. Väkevä and P. Ylöstalo (2021). Optical model for the Baltic Sea with an explicit CDOM state variable: a case study with Model ERGOM (version 1.2). Geosci. Model Dev. 14: 5049-5062, doi: 10.5194/gmd-14-5049-2021

V ANNEX

Example of structure and semantic of NetCDF files for product BALTICSEA_ANALYSISFORECAST_PHY_003_006/:

```
dimensions:
    time = UNLIMITED ; // (1 currently)
    depth = 56 ;
    lon = 763 ;
    lat = 774 ;
variables:
    double time(time) ;
        time:units = "days since 1900-01-01 00:00:00" ;
        time:long_name = "time" ;
        time:standard_name = "time" ;
        time:axis = "T" ;
    float depth(depth) ;
        depth:units = "m" ;
        depth:comment = "" ;
        depth:unit_long = "meters" ;
        depth:reference = "sea_level SeaDataNet L111" ;
        depth:positive = "down" ;
        depth:uncertainty = "" ;
        depth:valid_min = 0. ;
        depth:QC_procedure = 1 ;
        depth:long_name = "Depth of each measurement" ;
        depth:standard_name = "depth" ;
        depth:QC_indicator = 1 ;
        depth:valid_max = 12000. ;
        depth:axis = "Z" ;
    float lon(lon) ;
        lon:units = "degrees_east" ;
```

```
        lon:long_name = "Longitude of each location" ;
        lon:standard_name = "longitude" ;
        lon:unit_long = "degrees East" ;
        lon:axis = "X" ;
    float lat(lat) ;
        lat:units = "degrees_north" ;
        lat:long_name = "Latitude of each location" ;
        lat:standard_name = "latitude" ;
        lat:unit_long = "degrees North" ;
        lat:axis = "Y" ;
    float thetao(time, depth, lat, lon) ;
        thetao:_FillValue = -999.f ;
        thetao:units = "degree_Celsius" ;
        thetao:long_name = "potential temperature" ;
        thetao:standard_name = "sea_water_potential_temperature" ;
        thetao:unit_long = "degree Celsius" ;
        thetao:missing_value = -999.f ;
    float so(time, depth, lat, lon) ;
        so:_FillValue = -999.f ;
        so:units = "1e-3" ;
        so:long_name = "salinity" ;
        so:standard_name = "sea_water_salinity" ;
        so:unit_long = "1e-3" ;
        so:missing_value = -999.f ;
    float uo(time, depth, lat, lon) ;
        uo:_FillValue = -999.f ;
        uo:units = "m s-1" ;
        uo:long_name = "Eastward current" ;
        uo:standard_name = "eastward_sea_water_velocity" ;
        uo:unit_long = "meters per second" ;
        uo:missing_value = -999.f ;
    float vo(time, depth, lat, lon) ;
```

```
        vo:_FillValue = -999.f ;
        vo:units = "m s-1" ;
        vo:long_name = "Northward current" ;
        vo:standard_name = "northward_sea_water_velocity" ;
        vo:unit_long = "meters per second" ;
        vo:missing_value = -999.f ;
    float wo(time, depth, lat, lon) ;
        wo:_FillValue = -999.f ;
        wo:units = "m s-1" ;
        wo:long_name = "Upward current" ;
        wo:standard_name = "upward_sea_water_velocity" ;
        wo:unit_long = "meters per second" ;
        wo:missing_value = -999.f ;
    float bottomT(time, lat, lon) ;
        bottomT:_FillValue = -999.f ;
        bottomT:units = "degrees_C" ;
        bottomT:long_name = "Sea water potential temperature at sea floor (given for depth
comprise between 0 and 500m)" ;
        bottomT:standard_name = "sea_water_potential_temperature_at_sea_floor" ;
        bottomT:unit_long = "degree Celsius" ;
        bottomT:missing_value = -999.f ;
        bottomT:valid_range = -3.f, 40.f ;
    float sob(time, lat, lon) ;
        sob:_FillValue = -999.f ;
        sob:units = "0.001" ;
        sob:long_name = "Sea water salinity at sea floor" ;
        sob:standard_name = "sea_water_salinity_at_sea_floor" ;
        sob:unit_long = "0.001" ;
        sob:missing_value = -999.f ;
        sob:valid_range = 0.f, 40.f ;
  float sla(time, lat, lon) ;
        sla:_FillValue = -999.f ;
```

```
        sla:units = "m" ;
        sla:long_name = "Sea level elevation" ;
        sla:standard_name = "sea_surface_height_above_sea_level" ;
        sla:unit_long = "meters" ;
        sla:missing_value = -999.f ;
        sla:valid_range = -10.f, 10.f ;
    float sithick(time, lat, lon) ;
        sithick:_FillValue = -999.f ;
        sithick:units = "m" ;
        sithick:long_name = "Sea ice thickness" ;
        sithick:standard_name = "sea_ice_thickness" ;
        sithick:unit_long = "meters" ;
        sithick:missing_value = -999.f ;
    float siconc(time, lat, lon) ;
        siconc:_FillValue = -999.f ;
        siconc:units = "1" ;
        siconc:long_name = "Sea ice cover" ;
        siconc:standard_name = "sea_ice_area_fraction" ;
        siconc:missing_value = -999.f ;
        siconc:valid_range = 0.f, 1.f ;
    float mlotst(time, lat, lon) ;
        mlotst:_FillValue = -999.f ;
        mlotst:units = "m" ;
        mlotst:long_name = "Ocean mixed layer thickness defined by density (as in de Boyer
Montegut, 2004)" ;
        mlotst:standard_name = "ocean_mixed_layer_thickness_defined_by_sigma_theta" ;
        mlotst:unit_long = "meters" ;
        mlotst:missing_value = -999.f ;
```

Example of structure and semantic of NetCDF files for product BALTICSEA_ANALYSISFORECAST_BGC_003_007/:

```
    time = UNLIMITED ; // (1 currently)
    depth = 56 ;
    lon = 763 ;
    lat = 774 ;
variables:
    double time(time) ;
        time:units = "days since 1900-01-01 00:00:00" ;
        time:long_name = "time" ;
        time:standard_name = "time" ;
        time:axis = "T" ;
    float depth(depth) ;
        depth:units = "m" ;
        depth:comment = "" ;
        depth:unit_long = "meters" ;
        depth:reference = "sea_level SeaDataNet L111" ;
        depth:positive = "down" ;
        depth:uncertainty = "" ;
        depth:valid_min = 0. ;
        depth:QC_procedure = 1 ;
        depth:long_name = "Depth of each measurement" ;
        depth:standard_name = "depth" ;
        depth:QC_indicator = 1 ;
        depth:valid_max = 12000. ;
        depth:axis = "Z" ;
    float lon(lon) ;
        lon:units = "degrees_east" ;
        lon:long_name = "Longitude of each location" ;
        lon:standard_name = "longitude" ;
        lon:unit_long = "degrees East" ;
        lon:axis = "X" ;
```

```
    float lat(lat) ;
        lat:units = "degrees_north" ;
        lat:long_name = "Latitude of each location" ;
        lat:standard_name = "latitude" ;
        lat:unit_long = "degrees North" ;
        lat:axis = "Y" ;
    float chl(time, depth, lat, lon) ;
        chl:_FillValue = -999.f ;
        chl:units = "mg m-3" ;
        chl:long_name = "Chlorophyll_a_Concentration" ;
        chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
        chl:unit_long = "mass per unit volume" ;
        chl:missing_value = -999.f ;
    float phyc(time, depth, lat, lon) ;
        phyc:_FillValue = -999.f ;
        phyc:units = "mmol m-3" ;
        phyc:long_name = "Phytoplankton Concentration" ;
        phyc:standard_name =
"mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water" ;
        phyc:unit_long = "millimole C per unit volume" ;
        phyc:missing_value = -999.f ;
    float o2b(time, lat, lon) ;
        o2b:_FillValue = -999.f ;
        o2b:units = "mmol m-3" ;
        o2b:long_name = "Dissolved_Oxygen_Concentration" ;
        o2b:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
        o2b:unit_long = "millimole O2 per unit volume" ;
        o2b:missing_value = -999.f ;
    float no3(time, depth, lat, lon) ;
        no3:_FillValue = -999.f ;
        no3:units = "mmol m-3" ;
```

```
            no3:long_name = "Nitrate_Concentration" ;
            no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
            no3:unit_long = "millimole N per unit volume" ;
            no3:missing_value = -999.f ;
        float kd(time, depth, lat, lon) ;
            kd:_FillValue = -999.f ;
            kd:units = "1/m" ;
            kd:long_name = "Light Attenuation Coefficient" ;
            kd:standard_name =
"volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water" ;
            kd:unit_long = "1/m" ;
            kd:missing_value = -999.f ;
        float o2(time, depth, lat, lon) ;
            o2:_FillValue = -999.f ;
            o2:units = "mmol m-3" ;
            o2:long_name = "Dissolved_Oxygen_Concentration" ;
            o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
            o2:unit_long = "millimole O2 per unit volume" ;
            o2:missing_value = -999.f ;
        float pH(time, depth, lat, lon) ;
            pH:_FillValue = -999.f ;
            pH:units = "1" ;
            pH:long_name = "sea_water_ph" ;
            pH:standard_name = "sea_water_ph_reported_on_total_scale" ;
            pH:unit_long = "1" ;
            pH:missing_value = -999.f ;
    float po4(time, depth, lat, lon) ;
            po4:_FillValue = -999.f ;
            po4:units = "mmol m-3" ;
            po4:long_name = "Phosphate_Concentration" ;
            po4:standard_name = "mole_concentration_of_phosphate_in_sea_water" ;
```

```
            po4:unit_long = "millimole P per unit volume" ;
            po4:missing_value = -999.f ;
        float zooc(time, depth, lat, lon) ;
            zooc:_FillValue = -999.f ;
            zooc:units = "mmol m-3" ;
            zooc:long_name = "Zooplankton Concentration" ;
            zooc:standard_name =
"mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water" ;
            zooc:unit_long = "millimole C per unit volume" ;
            zooc:missing_value = -999.f ;
        float spCO2(time, lat, lon) ;
            spCO2:_FillValue = -999.f ;
            spCO2:units = "Pa" ;
            spCO2:long_name = "surface_partial_pressure_of_carbon_dioxide" ;
            spCO2:standard_name =
"surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
            spCO2:unit_long = "pascal" ;
            spCO2:missing_value = -999.f ;
        float nh4(time, depth, lat, lon) ;
            nh4:_FillValue = -999.f ;
            nh4:units = "mmol m-3" ;
            nh4:long_name = "Ammonium_Concentration" ;
            nh4:standard_name = "mole_concentration_of_ammonium_in_sea_water" ;
            nh4:unit_long = "millimole N per unit volume" ;
            nh4:missing_value = -999.f ;
        float dissic(time, depth, lat, lon) ;
            dissic:_FillValue = -999.f ;
            dissic:units = "mol m-3" ;
            dissic:long_name = "Dissolved_Inorganic_Carbon_Concentration" ;
            dissic:standard_name =
"mole_concentration_of_dissolved_inorganic_carbon_in_sea_water" ;
            dissic:unit_long = "millimole C per unit volume" ;
            dissic:missing_value = -999.f ;
```

```
    float si(time, depth, lat, lon) ;
        si:_FillValue = -999.f ;
        si:units = "mmol m-3" ;
        si:long_name = "Silica_Concentration" ;
        si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
        si:unit_long = "millimole Silicate per unit volume" ;
        si:missing_value = -999.f ;
    float fe(time, depth, lat, lon) ;
        fe:_FillValue = -999.f ;
        fe:units = "mmol m-3" ;
        fe:long_name = "mole_concentration_of_dissolved_iron_in_sea_water " ;
        fe:standard_name = "mole_concentration_of_dissolved_iron_in_sea_water" ;
        fe:unit_long = "millimole fe per unit volume" ;
        fe:missing_value = -999.f ;
    float cdom(time, depth, lat, lon) ;
        cdom:_FillValue = -999.f ;
        cdom:units = "mmol m-3" ;
         cdom:long_name = " mole_concentration_of_colored_dissolved_organic
_matter_expressed_as_carbon_in_sea_water " ;
        cdom:standard_name="mole_concentration_of_colored_dissolved_organic
_matter_expressed_as_carbon_in_sea_water" ;
        cdom:unit_long = "millimole cdom per unit volume" ;
        cdom:missing_value = -999.f ;
    float h2s(time, depth, lat, lon) ;
        h2s:_FillValue = -999.f ;
        h2s:units = "mmol m-3" ;
        h2s:long_name = " mole_concentration_of_hydrogen_sulfide_in_sea_water  " ;
        h2s:standard_name = "mole_concentration_of_hydrogen_sulfide_in_sea_water" ;
        h2s:unit_long = "millimole h2s per unit volume" ;
        h2s:missing_value = -999.f ;
```